

# MA388 Sabermetrics: Lesson 16

## Catcher Framing Ability Random Effects

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### Lesson Objectives

1. Discuss the limitations of a fixed-effects model with many indicators of a categorical variable.
2. Fit a random-effects (multilevel) model to estimate catcher-to-catcher variability.
3. Use likelihood ratio tests to compare nested models.
4. Interpret variance components to assess whether catchers meaningfully affect called-strike rates.

Last class, we were primarily comparing called strikes for Buster Posey and Yadier Molina. Today we'll look at more catchers from June 2019.

```
library(tidyverse)
library(knitr)
library(broom)
library(mgcv)

# Retrieve pitch level data from June 2019. (See previous lesson.)
pitches <- readRDS("statcast_june_2019.rds")

# Add catcher's name to the pitch data from the MLB master list.
mlbIDs <- baseballr::chadwick_player_lu() |>
  mutate(
    mlb_name = paste(name_first, name_last),
    mlb_id = key_mlbam
  )

pitches <- pitches |>
  left_join(select(mlbIDs, mlb_name, mlb_id),
            by = c("fielder_2" = "mlb_id")) |>
  rename(catcher_name = mlb_name)
```

```
#
# pitches_taken_subset <- pitches |>
#   filter(description %in% c("ball", "called_strike"))
```

## Random Effects Models

Thus far, we investigated the difference between Molina and Posey on called strikes. In doing so, we did the following:

1. Compared their called strike proportions (*why isn't this that useful?*)
2. Estimated the log odds ratio for a called strike after adjusting for pitch location (*why is this better?*)

**But, what we really want to know is not if Molina is better than Posey, but if catchers, in general, have a meaningful effect on called strikes.**

Based on what you've learned so far, you would probably fit a *fixed effects* model like this:

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 \text{catcher}_{1,i} + \dots + \beta_{m-1} \text{catcher}_{m-1,i} + f(\text{plate\_x}_i, \text{plate\_z}_i)$$

where  $\text{catcher}_{j,i}$  is an indicator of whether the catcher for pitch  $i$  is  $\text{catcher}_j$ .

How many parameters are there in the model above?

Let's fit the model for all catchers who caught at least 1000 pitches:

```
catcher_list <- pitches |>
  group_by(catcher_name) |>
  summarize(n = n()) |>
  filter(n >= 1000) |>
  pull(catcher_name)

pitches_taken <- pitches |>
  filter(catcher_name %in% catcher_list,
         description %in% c("called_strike", "ball")) |>
```

```
mutate(called_strike=description=='called_strike')

strike_mod_all <- gam(called_strike ~ s(plate_x, plate_z) + catcher_name,
                      family = "binomial", data = pitches_taken)

summary(strike_mod_all)
```

Family: binomial

Link function: logit

Formula:

called\_strike ~ s(plate\_x, plate\_z) + catcher\_name

Parametric coefficients:

|                               | Estimate  | Std. Error | z value | Pr(> z )   |
|-------------------------------|-----------|------------|---------|------------|
| (Intercept)                   | -6.622732 | 1.142689   | -5.796  | 6.8e-      |
| 09 ***                        |           |            |         |            |
| catcher_nameAustin Hedges     | 0.504288  | 0.202702   | 2.488   | 0.01285 *  |
| catcher_nameBobby Wilson      | 0.029825  | 0.245906   | 0.121   | 0.90346    |
| catcher_nameBrian McCann      | 0.056919  | 0.213540   | 0.267   | 0.78982    |
| catcher_nameBryan Holaday     | -0.256764 | 0.220585   | -1.164  | 0.24442    |
| catcher_nameBuster Posey      | -0.257745 | 0.215861   | -1.194  | 0.23247    |
| catcher_nameCam Gallagher     | 0.056798  | 0.241117   | 0.236   | 0.81377    |
| catcher_nameCarson Kelly      | 0.144979  | 0.199848   | 0.725   | 0.46818    |
| catcher_nameChance Sisco      | -0.473557 | 0.208069   | -2.276  | 0.02285 *  |
| catcher_nameChris Iannetta    | -0.390696 | 0.218313   | -1.790  | 0.07352 .  |
| catcher_nameChristian Vázquez | 0.443846  | 0.196247   | 2.262   | 0.02372 *  |
| catcher_nameCurt Casali       | -0.190821 | 0.205608   | -0.928  | 0.35337    |
| catcher_nameDanny Jansen      | 0.016908  | 0.198817   | 0.085   | 0.93223    |
| catcher_nameDustin Garneau    | -0.037663 | 0.242865   | -0.155  | 0.87676    |
| catcher_nameEliás Díaz        | -0.571133 | 0.191465   | -2.983  | 0.00285 ** |
| catcher_nameGary Sánchez      | 0.085060  | 0.192510   | 0.442   | 0.65860    |
| catcher_nameGrayson Greiner   | -0.337244 | 0.259916   | -1.298  | 0.19446    |
| catcher_nameJ. T. Realmuto    | -0.052478 | 0.192935   | -0.272  | 0.78562    |
| catcher_nameJames McCann      | -0.548860 | 0.200074   | -2.743  | 0.00608 ** |
| catcher_nameJason Castro      | -0.059725 | 0.208673   | -0.286  | 0.77471    |
| catcher_nameJeff Mathis       | -0.209227 | 0.204925   | -1.021  | 0.30726    |
| catcher_nameJohn Hicks        | -0.360805 | 0.218333   | -1.653  | 0.09842 .  |
| catcher_nameJonathan Lucroy   | -0.434742 | 0.197432   | -2.202  | 0.02767 *  |
| catcher_nameJorge Alfaro      | 0.302585  | 0.214719   | 1.409   | 0.15877    |
| catcher_nameJosh Phegley      | -0.418253 | 0.191059   | -2.189  | 0.02859 *  |
| catcher_nameKevin Plawecki    | 0.555562  | 0.250585   | 2.217   | 0.02662 *  |
| catcher_nameKurt Suzuki       | -0.230635 | 0.212513   | -1.085  | 0.27780    |
| catcher_nameLuke Maile        | -0.340531 | 0.219758   | -1.550  | 0.12124    |

|                               |           |          |        |         |    |
|-------------------------------|-----------|----------|--------|---------|----|
| catcher_nameManny Piña        | 0.532831  | 0.242093 | 2.201  | 0.02774 | *  |
| catcher_nameMartín Maldonado  | -0.099924 | 0.196022 | -0.510 | 0.61022 |    |
| catcher_nameMatt Wieters      | -0.330689 | 0.229847 | -1.439 | 0.15023 |    |
| catcher_nameMike Zunino       | -0.040333 | 0.206641 | -0.195 | 0.84525 |    |
| catcher_nameMitch Garver      | -0.025173 | 0.214782 | -0.117 | 0.90670 |    |
| catcher_nameOmar Narváez      | -0.390007 | 0.195070 | -1.999 | 0.04557 | *  |
| catcher_namePedro Severino    | -0.519941 | 0.201742 | -2.577 | 0.00996 | ** |
| catcher_nameRoberto Pérez     | 0.101593  | 0.198618 | 0.512  | 0.60900 |    |
| catcher_nameRobinson Chirinos | -0.317570 | 0.191141 | -1.661 | 0.09662 | .  |
| catcher_nameRussell Martin    | 0.224721  | 0.218288 | 1.029  | 0.30326 |    |
| catcher_nameSandy León        | 0.090762  | 0.222809 | 0.407  | 0.68375 |    |
| catcher_nameStephen Vogt      | -0.200628 | 0.217149 | -0.924 | 0.35553 |    |
| catcher_nameTim Federowicz    | 0.055533  | 0.220292 | 0.252  | 0.80097 |    |
| catcher_nameTom Murphy        | 0.251783  | 0.211580 | 1.190  | 0.23404 |    |
| catcher_nameTomás Nido        | 0.084321  | 0.228749 | 0.369  | 0.71241 |    |
| catcher_nameTony Wolters      | -0.300176 | 0.196645 | -1.526 | 0.12689 |    |
| catcher_nameTravis d'Arnaud   | -0.117914 | 0.214300 | -0.550 | 0.58216 |    |
| catcher_nameTucker Barnhart   | -0.002438 | 0.222311 | -0.011 | 0.99125 |    |
| catcher_nameTyler Flowers     | 0.436709  | 0.213844 | 2.042  | 0.04113 | *  |
| catcher_nameVíctor Caratini   | 0.111638  | 0.251220 | 0.444  | 0.65676 |    |
| catcher_nameWillson Contreras | -0.077984 | 0.191878 | -0.406 | 0.68443 |    |
| catcher_nameWilson Ramos      | -0.055399 | 0.196864 | -0.281 | 0.77840 |    |
| catcher_nameYadier Molina     | -0.106296 | 0.199854 | -0.532 | 0.59482 |    |
| catcher_nameYan Gomes         | -0.126179 | 0.209642 | -0.602 | 0.54725 |    |
| catcher_nameYasmani Grandal   | 0.156120  | 0.193919 | 0.805  | 0.42077 |    |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

|                    | edf   | Ref.df | Chi.sq | p-value    |
|--------------------|-------|--------|--------|------------|
| s(plate_x,plate_z) | 27.26 | 28.01  | 8062   | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.757 Deviance explained = 72.9%

UBRE = -0.65503 Scale est. = 1 n = 54411

What should we conclude from this model?

What are some limitations of a *fixed effects* approach for this question?

*Random effects* model the individual catcher effects as normally distributed with mean 0 and variance  $\sigma^2$ . There is only one parameter to fit. As a bonus, it has a nice interpretation...the larger the variance, the larger the effect of the catcher (*why?*). Here is the model:

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \text{catcher}_j + f(\text{plate\_x}_i, \text{plate\_z}_i)$$

where  $\text{catcher}_j$  is normally distributed with mean 0 and variance  $\sigma^2$ .

How many parameters are in this model?

Why is this an improvement?

We can incorporate this using the `lme4` package. Note, we'll do this slightly differently by first fitting the location model and then using the predictions from that model in a second model with the random effect.

```
model_location <- gam(called_strike ~ s(plate_x, plate_z),
                      family = "binomial",
                      data = pitches_taken)

pitches_taken <- model_location |>
  augment(type.predict = "response",
          newdata = pitches_taken) |>
  rename(strike_prob = .fitted)

# Now fit a random effects model adjusting for pitch location.
library(lme4)
model_random_effects_catcher <- glmer(called_strike ~ strike_prob + (1|catcher_name),
                                       family = "binomial",
                                       data = pitches_taken)
summary(model_random_effects_catcher)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: called_strike ~ strike_prob + (1 | catcher_name)
Data: pitches_taken
```

```
AIC          BIC      logLik -2*log(L)  df.resid
```

```

20889.2  20916.0  -10441.6  20883.2  54408

Scaled residuals:
    Min       1Q   Median       3Q      Max
-6.9648 -0.1583 -0.1373  0.1717  8.1197

Random effects:
    Groups             Name             Variance Std.Dev.
catcher_name (Intercept) 0.05715  0.2391
Number of obs: 54411, groups:  catcher_name, 53

Fixed effects:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.89584    0.04674  -83.36  <2e-16 ***
strike_prob  7.48119    0.05805  128.88  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
              (Intr)
strike_prob -0.572

```

```

# Get random effects for catchers.
catcher_effects_adj <- model_random_effects_catcher |>
  ranef() |>
  as_tibble() |>
  transmute(id = levels(grp), effect = condval) |>
  arrange(desc(effect))

catcher_effects_adj |> head(5)

```

```

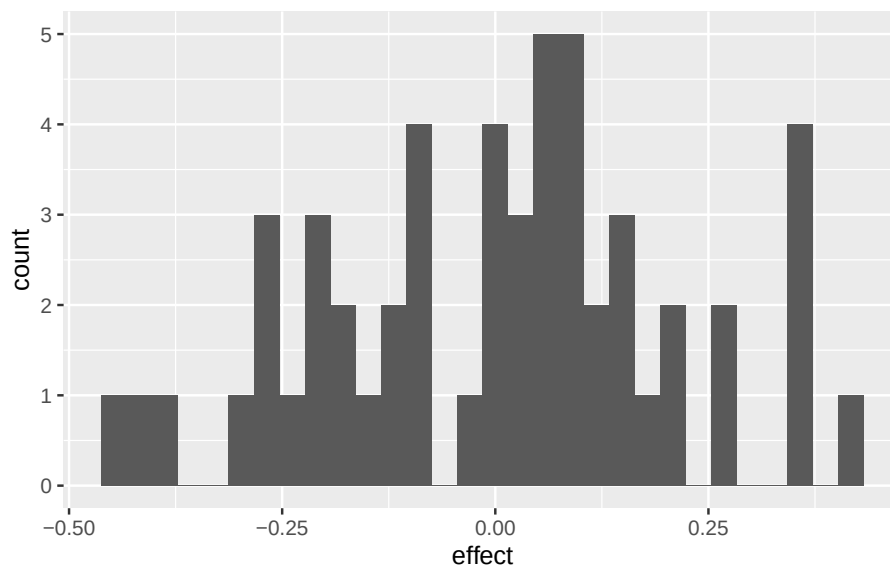
# A tibble: 5 x 2
  id          effect
<chr>        <dbl>
1 James McCann  0.432
2 Tony Wolters  0.366
3 Wilson Ramos  0.364
4 Jason Castro  0.359
5 Jorge Alfaro  0.343

```

```

catcher_effects_adj |>
  ggplot(aes(x = effect)) +
  geom_histogram()

```



Interesting - I guess? We're now faced with the same question we always have in statistics: I see a statistic, a non-zero difference, a non-zero variance, etc., but is it significant? What are some ways we can go about determining significance?

**Let's try this out!** Go to your AI of choice and ask the following question:

*I trained a random effects model using the glmer function in R and want to assess whether the random effect is significant or not. How might I do this?*

Let's use a nested model approach and add a couple more explanatory variables. Right now we've only let the catcher and the pitch location try to explain called strike determinations. If we introduce other variables that provide the same or complementary information, we might find the importance of catcher in the model to decrease. Sounds like ANOVA, right?

```
model_random_effects_trio <- glmer(called_strike ~ strike_prob +
                                   (1|pitcher) + (1|batter) + (1|catcher_name),
                                   family = "binomial",
                                   data = pitches_taken)
summary(model_random_effects_trio)
```

Generalized linear mixed model fit by maximum likelihood (Laplace

```

Approximation) [glmerMod]
Family: binomial ( logit )
Formula: called_strike ~ strike_prob + (1 | pitcher) + (1 | batter) +
(1 | catcher_name)
Data: pitches_taken

           AIC          BIC      logLik -2*log(L)  df.resid
20829.2    20873.7   -10409.6    20819.2     54406

Scaled residuals:
      Min       1Q   Median       3Q      Max
-8.0613 -0.1588 -0.1307  0.1659  8.8208

Random effects:
Groups          Name          Variance Std.Dev.
batter          (Intercept)  0.05828  0.2414
pitcher         (Intercept)  0.07996  0.2828
catcher_name    (Intercept)  0.04701  0.2168
Number of obs: 54411, groups:  batter, 600; pitcher, 531; catcher_name, 53

Fixed effects:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.98030    0.04928  -80.77   <2e-16 ***
strike_prob  7.63002    0.05965  127.92   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
              (Intr)
strike_prob -0.560

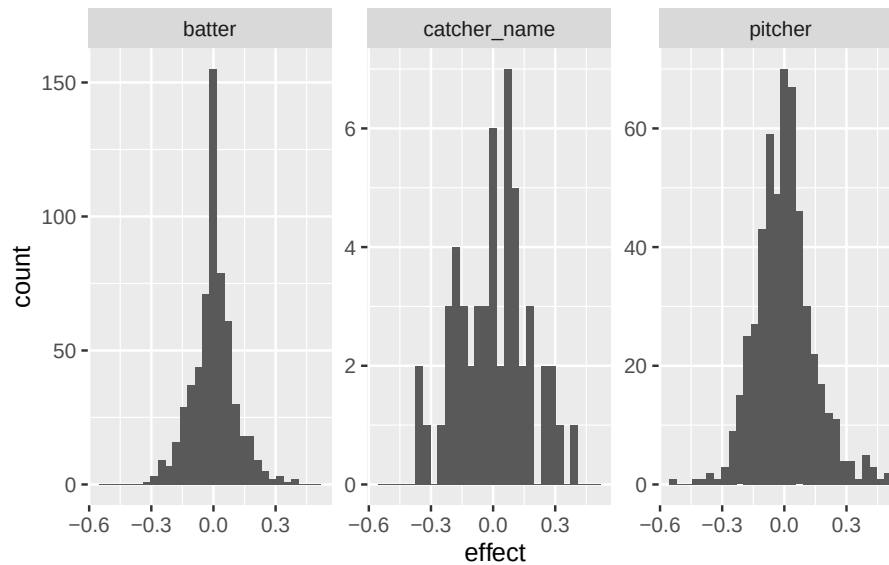
```

```

# Get random effects for pitchers, catchers, and batters.
player_effects_adj <- model_random_effects_trio |>
  ranef() |>
  as_tibble() |>
  mutate(id = grp, effect = condval) |>
  arrange(desc(effect))

player_effects_adj |>
  ggplot(aes(x = effect)) +
  geom_histogram() +
  facet_wrap(~grpvar, scales = "free_y")

```



```
model_random_effects_trio_no_catch <- glmer(called_strike ~ strike_prob + (1|pitcher) + (1|batter),
      family = "binomial",
      data = pitches_taken)

anova(model_random_effects_trio_no_catch, model_random_effects_trio, test = "LRT")
```

Data: pitches\_taken

Models:

model\_random\_effects\_trio\_no\_catch: called\_strike ~ strike\_prob + (1 | pitcher) + (1 | batter)

model\_random\_effects\_trio: called\_strike ~ strike\_prob + (1 | pitcher) + (1 | batter) + (1 | catcher\_name)

|                                    | npars | AIC   | BIC   | logLik | -2log(L) | Df     |
|------------------------------------|-------|-------|-------|--------|----------|--------|
| model_random_effects_trio_no_catch | 4     | 20860 | 20895 | -10426 |          | 20852  |
| model_random_effects_trio          | 5     | 20829 | 20874 | -10410 | 20819    | 32.362 |

2\*log(L) Chisq Df

model\_random\_effects\_trio\_no\_catch 4 20860 20895 -10426 20852

model\_random\_effects\_trio 5 20829 20874 -10410 20819 32.362 1

Pr(>Chisq)

model\_random\_effects\_trio\_no\_catch

model\_random\_effects\_trio 1.28e-08 \*\*\*

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1